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**REPUBLIC OF UZBEKISTAN
MINISTRY OF HIGHER EDUCATION, SCIENCE AND INNOVATIONS**

60610400 – Bachelor's Degree in Software Engineering

QUALIFICATION REQUIREMENT

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Tashkent – 2024

**60610600 – Bachelor's Degree in Software
Engineering**

DEVELOPED AND INTRODUCED:

**Tashkent Information Technologies University named after
Muhammad al-Khwarizmi.**

APPROVED AND IMPLEMENTED:

of the Ministry of Higher Education, Science and Innovation of the Republic
of Uzbekistan

By Order **No. 218** of June 25, 2024.

INTRODUCED:

Ministry of Higher Education, Science and Innovation of the Republic of
Uzbekistan.

These Qualification Requirements are based on the "State Educational Standard of Higher Education. General Provisions" and the "State Educational Standard of Higher Education. Classifier of Higher Education Fields and Specialties," the National and Sectoral Qualification Frameworks

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of the Republic of Uzbekistan, professional standards, and the proposals of personnel clients, and is an official normative-methodological document.

The official publication of Qualification Requirements within the territory of the Republic of Uzbekistan is the exclusive prerogative of the Ministry of Higher Education, Science and Innovation of the Republic of Uzbekistan.

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1. General Description

60610400 – Bachelor's training in the Software Engineering educational field is conducted in full-time and part-time (external) modes. Instruction for all modes of study is organized on a credit-module system. The standard duration of the bachelor's program for full-time students is 4 years.

1.1. Field of Application

1.1.1. Application of Qualification Requirements

60610400-The training of bachelor's degree students in the Software Engineering educational program represents a set of common requirements for all higher education institutions.

1.1.2. Main users of the qualification requirements:

- development and updating of the qualification requirements, curriculum, and study programs for this educational program, responsible for the effective implementation of the educational process and, within their authority, for the management staff of the higher education institution responsible for the graduates' level of preparation (rector, vice-rectors, head of the academic affairs department, deans and heads of departments) and its academic staff;
- students of a higher education institution who master the curriculum and course programs of the educational program;
- State Attestation Commissions that assess the preparedness level of bachelor's degree graduates;
- competent state bodies for education management;
- bodies that provide funding for higher education institutions;
- competent state bodies responsible for the accreditation and quality control of the higher education system;
- employer organizations and companies;
- Prospective students applying to higher education institutions, their parents, and other interested parties.

1.2. Description of Professional Activities

1.2.1. Fields of professional activity

state and non-state enterprises, organizations, institutions, and government bodies to solve complex software engineering issues in the field of information and communication technologies, the National Academy of Sciences of the Republic of Uzbekistan and branch scientific research institutes, This includes participation in research activities at scientific research centers, scientific-production associations, the Academy of Sciences of the Republic of Uzbekistan, and sectoral scientific-research institutes.

1.2.2. Objects of professional activities:

- mathematical and software foundations of automated and computer systems;
- automated design systems;

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- information processing, software for microprocessor systems;
- efficient processes for the implementation and use of software products and the hardware of computing equipment;
- processes of designing, configuring, manufacturing, and implementing information and communication systems and technologies, as well as their technical and software tools;
- Formulation of information exchange in electronic document circulation and databases in state interactive services and e-government systems;

Upon passing pedagogical retraining, graduates of the 60610400 Software Engineering bachelor's degree program are eligible to work in preschool and school education institutions, are entitled to engage in pedagogical activities teaching general professional and specialized disciplines in professional education institutions, as determined by the competent educational authorities.

1.2.3. Types of professional activities:

- organizational-management;
- design and development;
- operation and servicing;
- production-tuning;
- consulting services;

1.2.4. Professional Tasks

60610400-In accordance with the 6th Qualification Level of the National Qualifications Framework for the Software Engineering educational field, as well as the areas, objects, and types of bachelor's professional activities, a

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bachelor's graduate must be capable of performing the following professional tasks:

In organizational and management activities:

- developing methods and mechanisms for monitoring and assessing the quality of production processes related to software development and its operation;
- monitoring production processes for compliance with environmental protection and occupational safety requirements;
- developing and implementing software production quality management processes;
- quality management in the production process;
- organizing and managing the work of the team of performers;
- organizing and managing the work of primary design, technological, or production teams;
- Must be able to develop, execute, and monitor a work plan for the activities being carried out, and evaluate its results.

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In design and development activities:

- developing software, automated systems tools, and designing the project for testing and operation of software products;
- develop and research mathematical and simulation models for the subject of experimental and applied work being carried out;
- having the ability to develop design and software documentation;
- In operational activities, international and professional standards of information technology, modern paradigms and methodologies, as well as instrumental and computational tools, must be applied in accordance with the training specialty.

In the operation and provision of service activities:

- use and adjust software systems;
- must be capable of participating in the construction and installation of information and communication systems and networks and managing them with specialized tools.

In production and setup activities:

- developing and applying sample technological processes;
- planning the resources necessary to implement software production processes;
- ensuring environmental protection, fire, technical, and occupational safety in manufacturing processes;

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- developing and applying a system appropriate to the field in scientific and practical activities;
- planning and executing the assembly and commissioning of technical means, devices, transmission, storage, and processing systems of information and communication systems.

In consulting service activities:

- develop an analytical review of the existing situation in the field of software engineering;
- Consulting services must be provided on a topic that corresponds to the area of preparation.

2. Requirements for professional competencies

- be aware of the pressing issues of state policy and be able to independently analyze socio-economic problems and processes;
- understand the content of professional documents and materials in one of the foreign languages; possess the necessary knowledge in the natural sciences within the scope of their professional activity and be able to apply it in their professional practice on a modern scientific basis;
- be able to apply information technologies in their professional activities, have mastered the methods of collecting, storing, processing, and using information, and be able to make independent, well-founded decisions in their work;

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- be able to independently acquire new knowledge, work on self-improvement, and organize their professional activity on a scientific basis;
- have an understanding of the concept of a healthy lifestyle and the need to adhere to it.
- develop methods and mechanisms for monitoring and quality assessment of production processes related to software creation and its operation;
- applying methods for the analysis, synthesis, and optimization of programming processes, and product certification;
- applying mathematical models to information processing in software development;
- software development, development of algorithms and mathematical libraries;
- design and development of hardware-software systems;
- integration of software and firmware;

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- working as a junior researcher in scientific research organizations;
- teaching "Software Engineering" related subjects in accordance with established procedures in general secondary schools, academic lyceums, and technical schools;
- work in general secondary schools, academic lyceums, technical schools, and out-of-school institutions;
- Work in methodological, specialist, and other positions in the ministries responsible for education in the relevant fields, their branch offices, and institutions.

3. Requirements for Internships

Professional practice is aimed at consolidating theoretical knowledge from general professional and specialized disciplines and integrating it with practical (production) processes, as well as developing the corresponding practical skills, competencies, and qualifications.

A professional internship is conducted in the educational field.

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4. Structure of the Catalog of Sciences

T.r.	Course qualification code	Academic subjects, blocks, and types of activities	Total workload, in hours	Credit amount	Semester
1.00		Required courses	4500	150	1–7
1.01	UEY1104	The Latest History of Uzbekistan	120	4	1

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1.02	Religion 1204	Theology	120	4	2
1.03	PHIL1204	Philosophy	120	4	2
1.04	Foreign Language 1,2	Foreign Language 1,2	240	8	1.2
1.05	HIS1112	Calculus 1,2	360	12	1.2
1.06	PHYS1110	Physics 1,2	300	10	1,2
1.07	DIT1406	Differential Equations	180	6	4
1.08	CHA1306	Linear Algebra	180	6	3
1.09	DAS1110	Programming 1,2	300	10	1.2
1.10	AKY1104	Academic writing	120	4	1
1.11	MAB1306	Database	180	6	3
1.12	KIA1406	Cybersecurity Fundamentals	180	6	4
1.13	MTA1204	Data Structures and Algorithms	120	4	2
1.14	EVS1306	Electronics and Circuits 1	180	6	3
1.15	DIT1306	Discrete Structures	180	6	3
1.16	KOT1306	Computer Networks	180	6	3
1.17	SIA1406	Foundations of Artificial Intelligence	180	6	4
1.18	EVS1406	Probability and Statistics	180	6	4
1.19	INL1704	Individual project	120	4	7
1.20	ALL1506	Designing Algorithms	180	6	5
1.21	DTT1706	Software System Design	180	6	6
1.22	DIK1506	Introduction to Software Engineering	180	6	5
1.23	Operating Systems	Operating systems	180	6	5
1.24	DTT1704	Software testing	120	4	7
1.25	DTA1704	Software Architecture	120	4	7
2.00		Electives	1800	60	4–7

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		Total:	6300	210	1–7
Qualification		Engineer-Programmer			
Subject Code	Activity Type		Hour	Credit	Semester
MAA1830	Internship		900	30	8
Final State Certification	Final State Certification		–	–	8
	Total:		900	30	
	Total:		7200	240	

60610600 – Bachelor's Degree in Software Engineering**Bibliographic Information****UDC** 002: 651.1/.7**Group** T 55**OKS** 01.040.01**Keywords:**

Type of professional activity, competency, module, educational program, object of professional activity, field of professional activity, main bachelor's curriculum and course syllabus, law, regulation, decision, higher education, learning process, introduction to software engineering, algorithm design, software system design, programming styles and paradigms, operating systems, software architecture, software quality assurance, mobile application development, software requirements specification and analysis, graduation thesis, assessment, quality control, state accreditation, independent study, educational discipline block, table of contents, higher education institution, educational process, profile, practical training site, staff quality, workload, workload volume, internal control, final state examination, state and public oversight, external control, material and technical base.

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**Developer, agreed leading peer higher education institutions, and
personnel consumers**

DEVELOPED:

**Named after Muhammad al-Khwarizmi
Tashkent University of Information Technologies**

Rector _____ B. Maxhamov

M.O'.

June 14, 2024

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APPROVED:

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**General Director _____ M.
Maxmudov**

M.O'. June 20, 2024